

B.TECH 1ST YEAR · 1ST SEMESTER · ENGINEERING CHEMISTRY

Polymers & Fuel Chemistry

Unit III — Complete Formula & Concept Revision Sheet

Last-Day Revision · Exam-Ready · All Formulas + Mechanisms + Reactions

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DEFINITIONS

- ▶ **Polymer:** Large macromolecule made of repeating structural units (monomers) linked by covalent bonds.
- ▶ **Monomer:** Small, simple molecule that reacts to form a polymer. (e.g., ethylene → polyethylene)
- ▶ **Repeat Unit:** Smallest repeating structural unit in a polymer chain.

DEGREE OF POLYMERIZATION

$$n = M_{\text{polymer}} / M_{\text{monomer}}$$

- ▶ **n** = number of repeating units
- ▶ Higher **n** → higher molecular weight
- ▶ Typical **n**: $10^3 - 10^6$

FUNCTIONALITY OF MONOMERS

Monofunctional (f=1)

1 reactive group; chain terminator.
Forms no polymer alone. e.g., CH_3OH , acetic acid

Bifunctional (f=2)

2 reactive groups; forms linear polymers. e.g., ethylene glycol, adipic acid

Polyfunctional (f≥3)

≥3 reactive groups; forms branched/cross-linked polymers. e.g., glycerol (f=3), phenol (f=3)

CLASSIFICATION OF POLYMERS

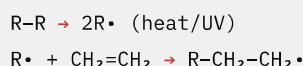
BASIS	TYPE	EXAMPLES
Source	Natural	Rubber, Cellulose, Starch, Silk, Wool
Source	Synthetic	Nylon, PVC, Polystyrene, Bakelite
Structure	Linear	HDPE, PVC, Nylon — high tensile strength
Structure	Branched	LDPE — lower density, flexible
Structure	Cross-linked / Network	Bakelite, Vulcanised rubber — rigid, thermoset
Polymerization	Addition (Chain-growth)	Polyethylene, PVC, Polystyrene
Polymerization	Condensation (Step-growth)	Nylon 6,6, Bakelite, Polyester



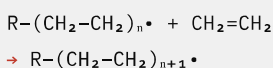
Polymerization Mechanisms

CHAIN GROWTH (ADDITION) POLYMERIZATION — FREE RADICAL MECHANISM

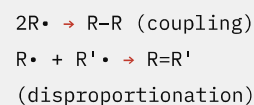
Step 1 – Initiation



Step 2 – Propagation

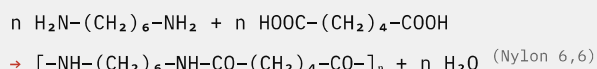


Step 3 – Termination



STEP GROWTH (CONDENSATION) POLYMERIZATION

- ▶ Monomers with **2 or more functional groups** react; small molecule (H_2O , HCl , CH_3OH) eliminated each step.
- ▶ Molecular weight builds **slowly**; high conversion needed for high MW.
- ▶ Any two molecules can react at any stage.



CHAIN GROWTH VS STEP GROWTH — COMPARISON

PROPERTY	CHAIN GROWTH	STEP GROWTH
Monomer type	Unsaturated compounds (C=C)	Bifunctional (–COOH, –OH, –NH ₂)
By-product	None (addition)	Small molecule (H ₂ O, HCl)
MW growth	Rapid (long chains form fast)	Slow (step-by-step)
Monomer consumption	Slow	Fast (early reaction)
Initiator needed?	Yes (free radical, ionic)	No (bifunctional monomers react)
Examples	PE, PVC, Polystyrene, Buna-S	Nylon 6,6, Bakelite, Polyester
MW of intermediates	Low (until termination)	Gradually increases

PROPERTY	THERMOPLASTICS	THERMOSETTING PLASTICS
Structure	Linear / Branched	Cross-linked / Network
On heating	Soften, can be remoulded	Harden permanently (char)
Recyclable?	✓ Yes	✗ No
Solubility	Soluble in organic solvents	Insoluble (cross-linked)
Strength	Lower	Higher (rigid)
Examples	PE, PVC, Polystyrene, Nylon	Bakelite, Urea formaldehyde, Epoxy
Polymerization	Chain / Step growth	Step growth (condensation)

KEY POLYMERS – PREPARATION, PROPERTIES & APPLICATIONS

◆ Polystyrene (PS) THERMOPLASTIC



PROPERTIES

- ▶ Transparent, rigid, brittle
- ▶ Good electrical insulator
- ▶ Low moisture absorption
- ▶ Easily moulded

APPLICATIONS

- ▶ Disposable cups, packaging foam
- ▶ Refrigerator linings
- ▶ Insulation (Thermocol)
- ▶ CD cases, toys

◆ Polyvinyl Chloride (PVC) THERMOPLASTIC



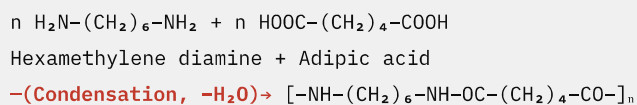
PROPERTIES

- ▶ Chemically resistant, tough
- ▶ Good dielectric properties
- ▶ Rigid (unplasticised) or flexible (plasticised)
- ▶ Flame retardant (Cl content)

APPLICATIONS

- ▶ Pipes, cables, wire insulation
- ▶ Floor tiles, garden hoses
- ▶ Raincoats, footwear
- ▶ Water tanks, bottles

◆ Nylon 6,6 THERMOPLASTIC CONDENSATION



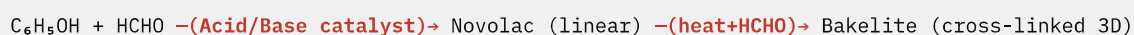
PROPERTIES

- ▶ High tensile strength & toughness
- ▶ Good abrasion resistance
- ▶ Self-lubricating
- ▶ Resistant to oil & chemicals
- ▶ High melting point (~265°C)

APPLICATIONS

- ▶ Textiles, stockings, ropes
- ▶ Gears, bearings, bushings
- ▶ Toothbrush bristles
- ▶ Parachute fabrics
- ▶ Fishing lines

◆ Bakelite (Phenol Formaldehyde) THERMOSET CONDENSATION



Mechanism: Phenol (f=3) + Formaldehyde → ortho/para substitution → methylene bridges → 3D network

PROPERTIES

- Hard, rigid, brittle
- Excellent electrical insulator
- Heat resistant, non-flammable
- Chemically inert

APPLICATIONS

- Electrical switches, sockets
- Brake linings, clutch plates
- Radio cabinets, handles
- Billiard balls

Definition: Polymers with elastic properties — can be stretched to several times their length and recover original shape on releasing stress. Cross-linking (vulcanisation) improves elasticity and durability.

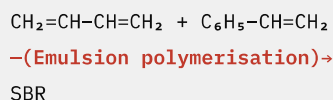
- ▶ **Key property:** Glass transition temperature (T_g) < room temperature → rubbery state
- ▶ **Vulcanisation:** S bridges between chains → reduces creep, improves tensile strength

ELASTOMERS COMPARISON

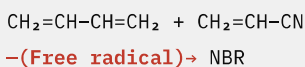
PROPERTY	BUNA-S (SBR)	BUNA-N (NBR)	THIOL (POLYSULPHIDE)
Monomers	Butadiene + Styrene	Butadiene + Acrylonitrile	Ethylene dichloride + Na_2S
Ratio	3:1 (B:S)	Varies (B:AN = 2:1)	—
Oil resistance	Poor	Excellent ✓	Excellent ✓
Heat resistance	Moderate	Good	Good
Elasticity	Good	Good	Moderate
Special feature	Abrasion resistant	Oil & fuel resistant	Solvent resistant
Applications	Tyres, footwear, belts	Gaskets, oil seals, hoses	Gaskets, aerospace sealants

PREPARATION REACTIONS

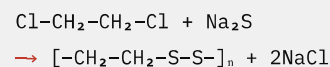
Buna-S



Buna-N



Thiokol



Fuels – Basics & Classification

Definition:

A fuel is a substance that on combustion in air/oxygen liberates heat energy economically and in a controllable manner.

CHARACTERISTICS OF GOOD FUEL

- ▶ High calorific value
- ▶ Low ignition temperature
- ▶ Low moisture & ash content
- ▶ Easy to store & transport
- ▶ Non-toxic combustion products
- ▶ Cheap & easily available

CLASSIFICATION OF FUELS

STATE	SOLID FUELS	LIQUID FUELS	GASEOUS FUELS
Natural	Wood, Coal, Peat, Lignite	Petroleum, Crude oil	Natural gas (CH_4)
Artificial	Coke, Charcoal, Briquettes	Petrol, Diesel, Kerosene	LPG, CNG, Biogas, Water gas
CV (approx)	25,000–35,000 kJ/kg	40,000–47,000 kJ/kg	35–55 MJ/m ³
Advantages	Abundant, cheap	High CV, easy handling	Cleaner burning, efficient

Higher (Gross) Calorific Value (HCV / GCV)

Heat liberated when 1 kg of fuel is completely burnt and combustion products are **cooled back to room temperature** (latent heat of water vapour is recovered).

Lower (Net) Calorific Value (LCV / NCV)

Heat liberated when 1 kg of fuel is completely burnt and water formed remains as **vapour** (latent heat NOT recovered).

Relationship:

$$\text{LCV} = \text{HCV} - (9H \times L) / 100$$

where H = % hydrogen in fuel, L = latent heat of steam $\approx 2465 \text{ kJ/kg}$ (587 kcal/kg)

$$\text{LCV} = \text{HCV} - 0.09 \times H \times 2465 \quad (\text{SI, kJ/kg})$$

DULONG'S FORMULA — HCV

$$\text{HCV} = 1/100 \times [33,800 C + 144,000 (H - O/8) + 9,270 S] \quad \text{kJ/kg}$$

- ▶ C = % Carbon in fuel
- ▶ H = % Hydrogen in fuel
- ▶ O = % Oxygen in fuel
- ▶ S = % Sulphur in fuel
- ▶ (H - O/8) = Available hydrogen (O already combined with H as water)

NUMERICAL PROBLEM FORMAT

Given: Fuel composition: C=%, H=%, O=%, S=%, N=%, Moisture=%

Step 1: Check composition sums to ~100%

Step 2: Apply Dulong's formula → HCV

Step 3: Apply $\text{LCV} = \text{HCV} - (9H/100) \times 2465 \text{ kJ/kg}$

Units:

- ▶ SI: kJ/kg or MJ/kg
- ▶ CGS: kcal/kg
- ▶ Conversion: 1 kcal = 4.187 kJ
- ▶ 1 MJ = 1000 kJ

BOMB CALORIMETER — EXPERIMENTAL HCV

$$\text{HCV} = (W + w) \times \Delta T \times 4.187 / m \quad \text{kJ/kg}$$

- ▶ W = water equivalent of calorimeter (g)
- ▶ w = mass of water taken (g)
- ▶ ΔT = rise in temperature ($^{\circ}\text{C}$)
- ▶ m = mass of fuel burned (g)
- ▶ Bomb calorimeter measures **HCV** (constant volume)

BOY'S CALORIMETER — EXPERIMENTAL LCV (GASEOUS FUELS)

- Used for gaseous fuels (coal gas, LPG)
- Water vapour passes out with exhaust → measures LCV directly
- **Formula:** LCV (in kJ/m³) = [Heat absorbed by water] / [Volume of gas burnt]

PROXIMATE ANALYSIS

Determines: Moisture, Volatile Matter, Ash, Fixed Carbon

PARAMETER	PROCEDURE	SIGNIFICANCE
Moisture (M)	Heat 1g coal at 105–110°C for 1hr; cool & weigh. Loss in mass = moisture.	High M → low CV, difficult ignition, increases transport cost
Volatile Matter (VM)	Heat residue at 950°C for 7 min in covered crucible. Loss in mass – moisture = VM.	High VM → easy ignition, long smoky flame, used for gas production
Ash (A)	Burn residue at 700–750°C till constant mass. Remaining = Ash.	High ash → low CV, slag, clinkering; increases handling cost
Fixed Carbon (FC)	$FC = 100 - (M + VM + A)$	High FC → high CV, burns with short blue flame; good coking coal

ULTIMATE ANALYSIS

Determines: C, H, N, S, O (by difference) & Ash

More accurate; used for Dulong's formula calculation

ELEMENT	SIGNIFICANCE
Carbon (C)	Main combustible; higher C → higher CV
Hydrogen (H)	High CV (144 MJ/kg); forms H ₂ O on combustion → reduces LCV
Nitrogen (N)	Inert; does not contribute to CV; high N → NO _x emissions
Sulphur (S)	Combustible (small CV); SO ₂ on combustion → acid rain, corrosion
Oxygen (O)	Inert (already oxidised); reduces available H; $O = 100 - (C + H + N + S + \text{Ash})$
Ash	Non-combustible; reduces CV, causes slagging

PROXIMATE VS ULTIMATE ANALYSIS — QUICK COMPARISON

BASIS	PROXIMATE ANALYSIS	ULTIMATE ANALYSIS
Purpose	Practical quality assessment	Theoretical/accurate CV calculation
Parameters	M, VM, Ash, FC	C, H, N, S, O, Ash
Equipment	Muffle furnace, balance	Combustion apparatus (C, H by Liebig; N by Kjeldahl)
Cost	Simple & cheap	Complex & expensive
Use	Coal grading, plant control	Dulong's formula, stoichiometry

COKE NUMBER / FUEL RANKING (QUALITY SCALE)

COAL TYPE	CARBON %	CV (KJ/KG)	USE
Anthracite	90–95%	~34,000	Domestic heating
Bituminous	75–90%	~30,000–33,000	Coke making, steam plants
Lignite (Brown coal)	25–75%	~15,000–25,000	Power stations
Peat	<25%	<15,000	Domestic (damp regions)

Formula recall:

$$\text{Fixed Carbon \%} = 100 - (\text{Moisture\%} + \text{Volatile Matter\%} + \text{Ash\%})$$

$$\text{Oxygen\% (Ultimate)} = 100 - (C + H + N + S + \text{Ash})\%$$

Petroleum (Crude Oil): Complex mixture of hydrocarbons (alkanes, cycloalkanes, aromatics) + small amounts of S, N, O compounds. Formed by decomposition of ancient marine organisms over millions of years under heat & pressure.

FRACTIONAL DISTILLATION — PRODUCTS & BOILING RANGES

FRACTION	BP RANGE (°C)	CARBON ATOMS	USES
Petroleum Gas (LPG)	<30°C	C ₁ –C ₄	Cooking fuel, heating
Petroleum Ether / Naphtha	30–90°C	C ₅ –C ₇	Solvent, dry cleaning
Gasoline (Petrol)	40–200°C	C ₅ –C ₁₂	Motor fuel (SI engines)
Kerosene (Paraffin)	150–300°C	C ₁₂ –C ₁₅	Jet fuel, lamps, stoves
Diesel / Gas Oil	250–350°C	C ₁₅ –C ₁₈	Diesel engines, furnaces
Lubricating Oil	350–400°C	C ₁₆ –C ₂₀	Lubrication
Paraffin Wax	400–500°C	C ₂₀ –C ₃₀	Candles, polish, waterproofing
Bitumen / Asphalt	Residue	>C ₃₀	Road surfacing, roofing

CRACKING — INCREASING PETROL YIELD

Thermal Cracking

- T = 450–750°C, high pressure
- Breaks C–C bonds randomly
- Produces straight-chain alkanes & alkenes
- Low octane number products

Catalytic Cracking (FCC)

- T = 450–550°C, zeolite catalyst
- Produces branched & aromatic hydrocarbons
- Higher octane number ✓
- More economical



Octane Number & Cetane Number

Octane Number (ON)

Measure of **anti-knock** property of SI engine fuels (petrol). % of iso-octane in a mixture of iso-octane + n-heptane that has the **same knocking tendency** as the test fuel.

- Iso-octane (2,2,4-trimethylpentane) → ON = **100**
- n-Heptane → ON = **0**
- Higher ON → less knocking → better fuel
- Modern petrol: ON = 87–95
- **Anti-knock agents:** TEL (tetraethyl lead), MTBE, ethanol

Cetane Number (CN)

Measure of **ignition quality** (ease of self-ignition) of CI engine fuels (diesel). % of cetane in a mixture of cetane + α-methylnaphthalene that has the **same ignition delay** as the test fuel.

- Cetane (n-hexadecane, C₁₆H₃₄) → CN = **100**
- α-Methylnaphthalene → CN = **0**
- Higher CN → shorter ignition delay → better diesel
- Good diesel: CN = 45–65
- **Cetane improvers:** ethyl nitrate, amyl nitrate

OCTANE VS CETANE — QUICK COMPARISON

PARAMETER	OCTANE NUMBER	CETANE NUMBER
For engine type	SI (petrol/spark ignition)	CI (diesel/compression ignition)
Measures	Anti-knock property	Ignition quality / ease of ignition
Reference fuels	Iso-octane (100) + n-Heptane (0)	Cetane (100) + α-Methylnaphthalene (0)
Ideal value	High ON preferred	High CN preferred
Branched alkanes	High ON (good for petrol)	Low CN (bad for diesel)

Straight-chain alkanes	Low ON	High CN (good for diesel)
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Key Insight: Branched hydrocarbons → ↑ ON (good petrol) but ↓ CN (bad diesel). Straight-chain hydrocarbons → ↓ ON but ↑ CN. **Opposite trends!**

FUEL	SOURCE	CV	ADVANTAGES	DISADVANTAGES	APPLICATIONS
Propane (LPG)	Petroleum refining, natural gas	~50.3 MJ/kg	Clean burning, high CV, easy liquefaction	Non-renewable, leak risk (heavier than air)	Cooking, vehicles (autogas), heating
Methanol (CH₃OH)	Synthesis gas (CO + H ₂), biomass	~22.7 MJ/kg	Biodegradable, high ON (107), burns cleaner, low NO _x	Toxic, corrosive, low CV vs petrol, absorbs water	Racing fuel, fuel cells, biodiesel feedstock
Ethanol (C₂H₅OH)	Fermentation of sugarcane, corn	~29.6 MJ/kg	Renewable, ON=111, reduces GHG emissions, biodegradable	Absorbs water, lower CV vs petrol, land use	E10/E85 blends with petrol, racing fuel, disinfectant
Biodiesel (FAME)	Transesterification of vegetable oils/animal fats + methanol	~37 MJ/kg	Renewable, biodegradable, high CN (~55), reduces PM & CO	High cloud point (gels in cold), may need blending	B5/B20/B100 blends in diesel engines
CNG (CH₄)	Natural gas, compressed to 200 bar	~53 MJ/kg	Cleanest fossil fuel, low CO ₂ , high ON	Bulky storage tank, limited infrastructure	Public transport, taxis, buses
Hydrogen (H₂)	Electrolysis, steam reforming	~142 MJ/kg	Highest CV, zero carbon emissions, water byproduct only	Expensive production, storage challenges, explosive	Fuel cells, space rockets, future vehicles

BIODIESEL PRODUCTION — TRANSESTERIFICATION REACTION

Triglyceride (vegetable oil) + 3 CH₃OH $\xrightarrow{\text{(NaOH catalyst)}}$ 3 FAME (biodiesel) + Glycerol
FAME = Fatty Acid Methyl Ester

ENVIRONMENTAL IMPACT COMPARISON

FUEL	CO ₂	NO _x	CO	SO _x	PARTICULATES
Petrol/Diesel (conventional)	High	High	High	Medium	Medium–High
CNG	Low	Low	Very Low	Very Low	Very Low
Ethanol blend (E85)	Lower	Similar	Lower	Very Low	Low
Biodiesel (B100)	Near zero (net)	Slightly higher	Lower	Very Low	Lower
Hydrogen	Zero	Low	Zero	Zero	Zero

CONVENTIONAL VS ALTERNATIVE FUELS

BASIS	CONVENTIONAL FUELS	ALTERNATIVE FUELS
Source	Fossil fuels (finite)	Renewable / synthetic
Emissions	High GHG, SO _x , NO _x	Lower / zero emissions
Availability	Declining reserves	Sustainable long-term
Infrastructure	Established	Developing
Cost	Established market price	Often higher initially
Energy density	High (petrol, diesel)	Often lower (ethanol, biodiesel)

Q ▶ Definition of a polymer? A ▶ Large macromolecule formed by repetitive bonding of monomers; n (degree of polymerisation) = $M_{\text{polymer}} / M_{\text{monomer}}$	Q ▶ Functionality of a monomer? A ▶ Number of reactive bonding sites (functional groups) available for polymerisation
Q ▶ Steps in chain growth (free radical) polymerisation? A ▶ 1. Initiation 2. Propagation 3. Termination	Q ▶ By-product in condensation polymerisation? A ▶ Small molecule eliminated: H_2O, HCl, or CH_3OH in each step
Q ▶ Difference: thermoplastic vs thermoset? A ▶ Thermoplastics soften on heating (recyclable); thermosets harden permanently (non-recyclable)	Q ▶ Monomers of Nylon 6,6? A ▶ Hexamethylene diamine $[H_2N(CH_2)_6NH_2]$ + Adipic acid $[HOOC(CH_2)_4COOH]$
Q ▶ Monomers of Bakelite? A ▶ Phenol (C_6H_5OH) + Formaldehyde ($HCHO$) in 3D cross-linked network	Q ▶ Why is Bakelite a thermoset? A ▶ Phenol has $f=3 \rightarrow$ forms 3D cross-linked network; cannot be remelted
Q ▶ Buna-S monomers & ratio? A ▶ Butadiene + Styrene in 3:1 ratio; made by emulsion polymerisation	Q ▶ Which elastomer is oil resistant? A ▶ Buna-N (NBR) — acrylonitrile group provides excellent oil & fuel resistance
Q ▶ Thiokol monomers? A ▶ Ethylene dichloride ($ClCH_2CH_2Cl$) + Sodium polysulphide (Na_2S_x)	Q ▶ What is vulcanisation? A ▶ Cross-linking rubber with sulphur ($140-160^\circ C$) to form $-S-S-$ bridges $\rightarrow$ improves elasticity, tensile strength, reduces creep
Q ▶ Define HCV? A ▶ Heat released when 1kg fuel is fully burnt AND combustion products cooled to room temp (latent heat of water included)	Q ▶ LCV formula? A ▶ $LCV = HCV - 0.09 \times H\% \times 2465 \text{ kJ/kg}$ (H = hydrogen percentage)
Q ▶ Dulong's formula? A ▶ $HCV = 1/100 \times [33,800C + 144,000(H - O/8) + 9,270S]$ kJ/kg	Q ▶ Why subtract O/8 from H in Dulong's formula? A ▶ O already combined with H as water internally $\rightarrow$ H unavailable for combustion = $(H - O/8)$ = available H
Q ▶ Proximate analysis gives? A ▶ Moisture (M), Volatile Matter (VM), Ash (A), Fixed Carbon ($FC = 100 - M - VM - A$)	Q ▶ High fixed carbon coal $\rightarrow$? A ▶ High CV, short blue flame, good for coking (anthracite)
Q ▶ High volatile matter coal $\rightarrow$? A ▶ Easy ignition, long smoky flame, better for gas production	Q ▶ What does Sulphur in coal do? A ▶ Burns to $SO_2 \rightarrow$ acid rain, corrosion of equipment; reduces coal quality
Q ▶ Octane number indicates? A ▶ Anti-knock property of SI (petrol) engine fuel; % iso-octane in iso-octane/n-heptane reference mixture	Q ▶ Cetane number indicates? A ▶ Ignition quality (ease of self-ignition) of CI (diesel) fuel; % cetane in cetane/α-methyl-naphthalene mixture
Q ▶ iso-Octane ON & CN values? A ▶ ON = 100 (best anti-knock); CN = very low (poor diesel fuel)	Q ▶ Fractions from petroleum distillation in order? A ▶ LPG $\rightarrow$ Naphtha $\rightarrow$ Petrol $\rightarrow$ Kerosene $\rightarrow$ Diesel $\rightarrow$ Lube oil $\rightarrow$ Paraffin wax $\rightarrow$ Bitumen ($\uparrow$ BP)

Q ▸ Cracking of petroleum — purpose?

A ▸ Breaks large, heavy HC molecules into smaller, lighter ones (petrol fraction) to increase yield of useful fuels

Q ▸ Biodiesel production reaction?

A ▸ Transesterification: Triglyceride + $3\text{CH}_3\text{OH} \rightarrow 3 \text{ FAME}$ (biodiesel) + Glycerol (NaOH catalyst)

Q ▸ Advantage of ethanol as fuel?

A ▸ Renewable (fermentation), high ON (111), biodegradable, reduces CO_2 emissions

Q ▸ PVC monomer & catalyst?

A ▸ Vinyl chloride ($\text{CH}_2=\text{CHCl}$); free radical catalyst (peroxide / benzoyl peroxide)

Q ▸ Which instrument measures HCV experimentally?

A ▸ Bomb calorimeter (constant volume); $\text{HCV} = (W+w) \times \Delta T \times 4.187/\text{m kJ/kg}$

Q ▸ Why does CNG have lower CO_2 emissions?

A ▸ CH_4 has highest H:C ratio among hydrocarbons $\rightarrow$ more H_2O formed relative to CO_2 per unit energy



-
- MISTAKE** **Confusing chain growth vs step growth:** Chain growth requires an initiator (free radical) and has fast chain formation with slow monomer loss. Step growth has NO initiator; ANY two molecules react; MW builds slowly. Remember: **Chain = fast chains, slow monomer loss | Step = fast monomer loss, slow MW build**
-
- MISTAKE** **Missing termination step in free radical mechanism:** Always write ALL 3 steps — Initiation (2 radicals from initiator), Propagation (chain extension), Termination (coupling OR disproportionation). Incomplete answer → lose marks.
-
- UNITS** **Forgetting units in calorific value numericals:** HCV must be in kJ/kg (SI). If Dulong's gives kcal, convert: $\times 4.187$ to get kJ. LCV formula uses $L = 2465 \text{ kJ/kg}$ (or 587 kcal/kg). State units explicitly at every step.
-
- FORMULA** **Errors in Dulong's formula:** The term is **$(H - O/8)$** , NOT just H. If $O > 8H$, the available H term becomes zero (never negative!). Also 33,800 for C; 144,000 for H; 9,270 for S — memorise these constants precisely.
-
- MISTAKE** **Confusing Octane and Cetane Numbers:** ON = anti-knock = for PETROL (SI engine). CN = ignition quality = for DIESEL (CI engine). High ON fuel $\neq$ good diesel fuel. **Branched HC → high ON but low CN. Straight-chain HC → low ON but high CN.**
-
- MISTAKE** **Skipping significance in coal analysis:** Exam questions often ask "significance of each constituent." Moisture → lowers CV & increases cost. VM → ignition ease. Ash → slagging & reduces CV. FC → actual heating value. Sulphur → SO_2 & acid rain.
-
- MISTAKE** **Mixing thermoplastics and thermosets:** Key test: Can it be remoulded? Yes → thermoplastic (linear structure). No → thermoset (cross-linked network). Bakelite is thermoset despite containing carbon rings. Nylon is thermoplastic despite condensation mechanism.
-
- FORMULA** **Fixed carbon formula:** $\text{FC} = 100 - (\text{M} + \text{VM} + \text{A})$. Students often forget to subtract ALL three. If question gives only three values, the fourth is 100 minus the rest.
-
- NOTE** **Nylon 6 vs Nylon 6,6:** Nylon 6,6 → from two different monomers (diamine + diacid). Nylon 6 → from ONE monomer (caprolactam) by ring-opening polymerisation. If not specified, exam likely means Nylon 6,6.
-
- NOTE** **Buna-N vs Buna-S oil resistance:** Only Buna-N (NBR) is oil resistant. The acrylonitrile ($-\text{CN}$) group is responsible. Buna-S (SBR) has poor oil resistance but good abrasion resistance (used in tyres).
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- MISTAKE** **LCV > HCV error:** LCV is ALWAYS less than HCV because water vapour latent heat is NOT recovered. If calculation gives $\text{LCV} > \text{HCV}$, recheck: sign error or wrong formula applied.
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- NOTE** **Bomb calorimeter measures HCV** (constant volume, water vapour condenses). Boy's calorimeter measures LCV for gaseous fuels (water exits as vapour). Do not mix them up.
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- UNITS** **Petroleum fractions order:** Low BP → small carbon number. Memorise sequence: LPG ($< \text{C}_4$) → Petrol ($\text{C}_5\text{--C}_{12}$) → Kerosene ($\text{C}_{12}\text{--C}_{15}$) → Diesel ($\text{C}_{15}\text{--C}_{18}$) → Lube oil → Bitumen. If sequence is jumbled in exam, ascending BP order = ascending carbon chain length.
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- MISTAKE** **Biodiesel vs Biogas:** Biodiesel = FAME (transesterification of oils). Biogas = $\text{CH}_4 + \text{CO}_2$ (anaerobic digestion of organic waste). Different processes, different fuels — don't confuse them.
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CORE FORMULAS

Degree of Polymerisation	$n = M_{\text{polymer}} / M_{\text{monomer}}$
HCV (Dulong's)	$[33800C + 144000(H - O/8) + 9270S] / 100 \text{ kJ/kg}$
LCV from HCV	$HCV - 0.09 \times H \times 2465 \text{ kJ/kg}$
Bomb Calorimeter HCV	$(W+w) \times \Delta T \times 4.187 / m \text{ kJ/kg}$
Fixed Carbon %	$100 - (M\% + VM\% + Ash\%)$
Oxygen % (Ultimate)	$100 - (C + H + N + S + Ash)\%$
LCV ↔ HCV relation	LCV < HCV always
Unit conversion	$1 \text{ kcal} = 4.187 \text{ kJ} \cdot 1 \text{ MJ} = 1000 \text{ kJ}$

KEY REACTIONS

Polystyrene	$n \text{ CH}_2=\text{CH}(\text{C}_6\text{H}_5) \rightarrow [-\text{CH}_2-\text{CH}(\text{C}_6\text{H}_5)-]_n$
PVC	$n \text{ CH}_2=\text{CHCl} \rightarrow [-\text{CH}_2-\text{CHCl}-]_n$
Nylon 6,6	diamine + adipic acid → polyamide + H ₂ O
Bakelite	phenol + HCHO → 3D cross-linked (acid/base cat.)
Buna-S	butadiene + styrene (3:1) → SBR
Buna-N	butadiene + acrylonitrile → NBR
Thiokol	$\text{ClCH}_2\text{CH}_2\text{Cl} + \text{Na}_2\text{S} \rightarrow [\text{CH}_2\text{CH}_2\text{SS}]_n + 2\text{NaCl}$
Biodiesel (transest.)	Triglyceride + 3CH ₃ OH → 3FAME + Glycerol

KEY CONSTANTS & REFERENCE VALUES

Latent heat of steam (L)	2465 kJ/kg = 587 kcal/kg
CV of C	33,800 kJ/kg
CV of H	144,000 kJ/kg
CV of S	9,270 kJ/kg
iso-Octane ON	100 (reference)
n-Heptane ON	0 (reference)
Cetane CN	100 (reference)
α-Methylnaphthalene CN	0 (reference)
Petrol ON (typical)	87–95
Diesel CN (typical)	45–65
Ethanol ON	111
H ₂ CV (highest)	~142 MJ/kg

CRITICAL KEYWORDS

Initiation	$R-R \rightarrow 2R\cdot$ (free radical)
Propagation	$R\cdot + \text{monomer} \rightarrow \text{extended chain}\cdot$
Termination	coupling / disproportionation
Vulcanisation	S bridges in rubber → ↑ strength
Cracking	large HC → small HC (↑ petrol yield)
FCC	Fluid Catalytic Cracking (zeolite cat.)
FAME	Fatty Acid Methyl Ester = Biodiesel
Available H	$(H - O/8)$ in Dulong's formula

POLYMER PROPERTIES SUMMARY

POLYMER	TYPE	MECHANISM	SPECIAL
Polystyrene	TP	Chain	Brittle, insulator
PVC	TP	Chain	Flame retardant
Nylon 6,6	TP	Step	High tensile
Bakelite	TS	Step	3D cross-linked
Buna-S (SBR)	Elastomer	Chain	Abrasion resistant
Buna-N (NBR)	Elastomer	Chain	Oil resistant
Thiokol	Elastomer	Step	Solvent resistant

TP = Thermoplastic · TS = Thermosetting

